

Drugs used in infertility and assisted reproduction

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Infertility : Failure to conceive after one year of regular unprotected sex.

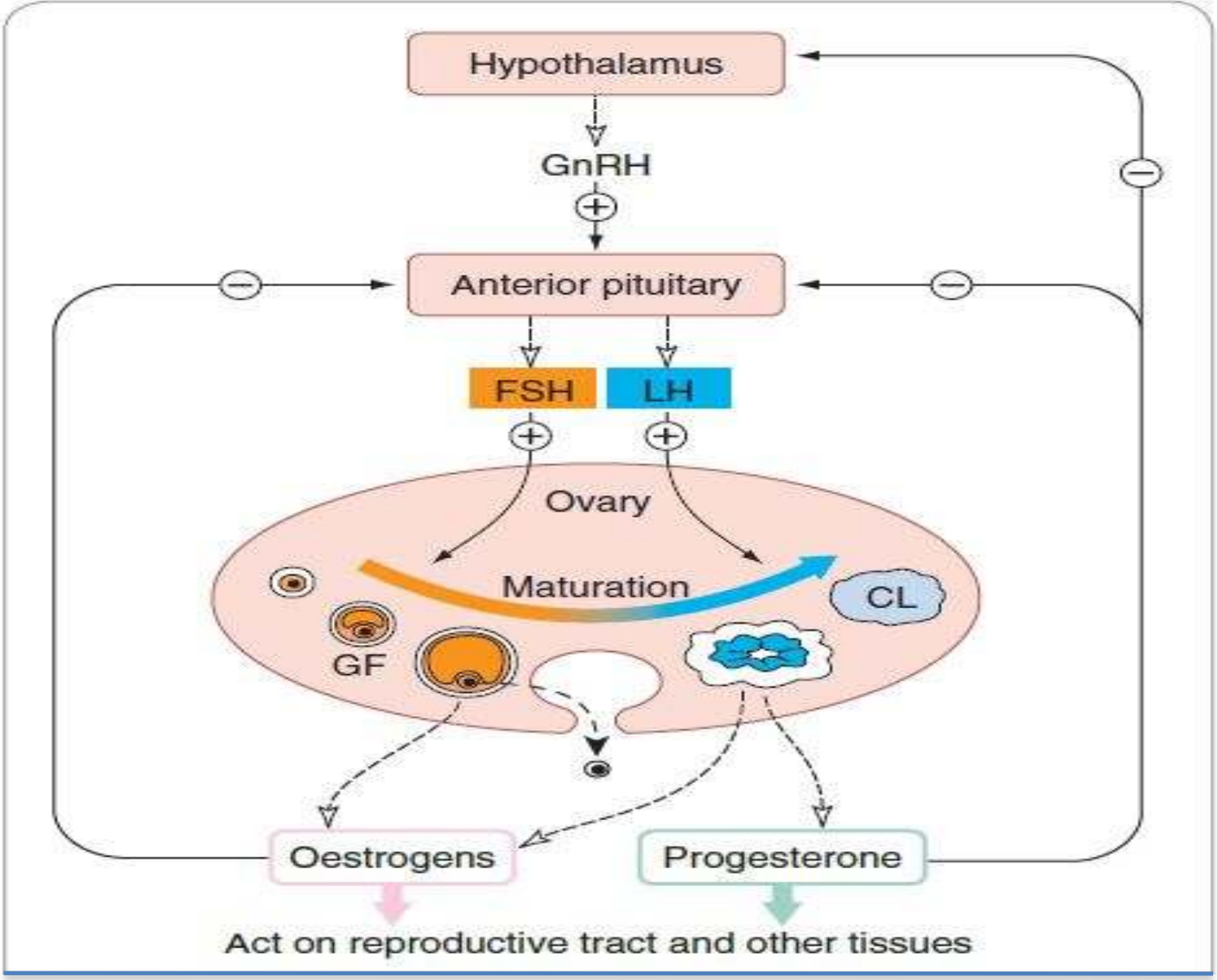
88% to 90% of couples will conceive within 1 year

Female:

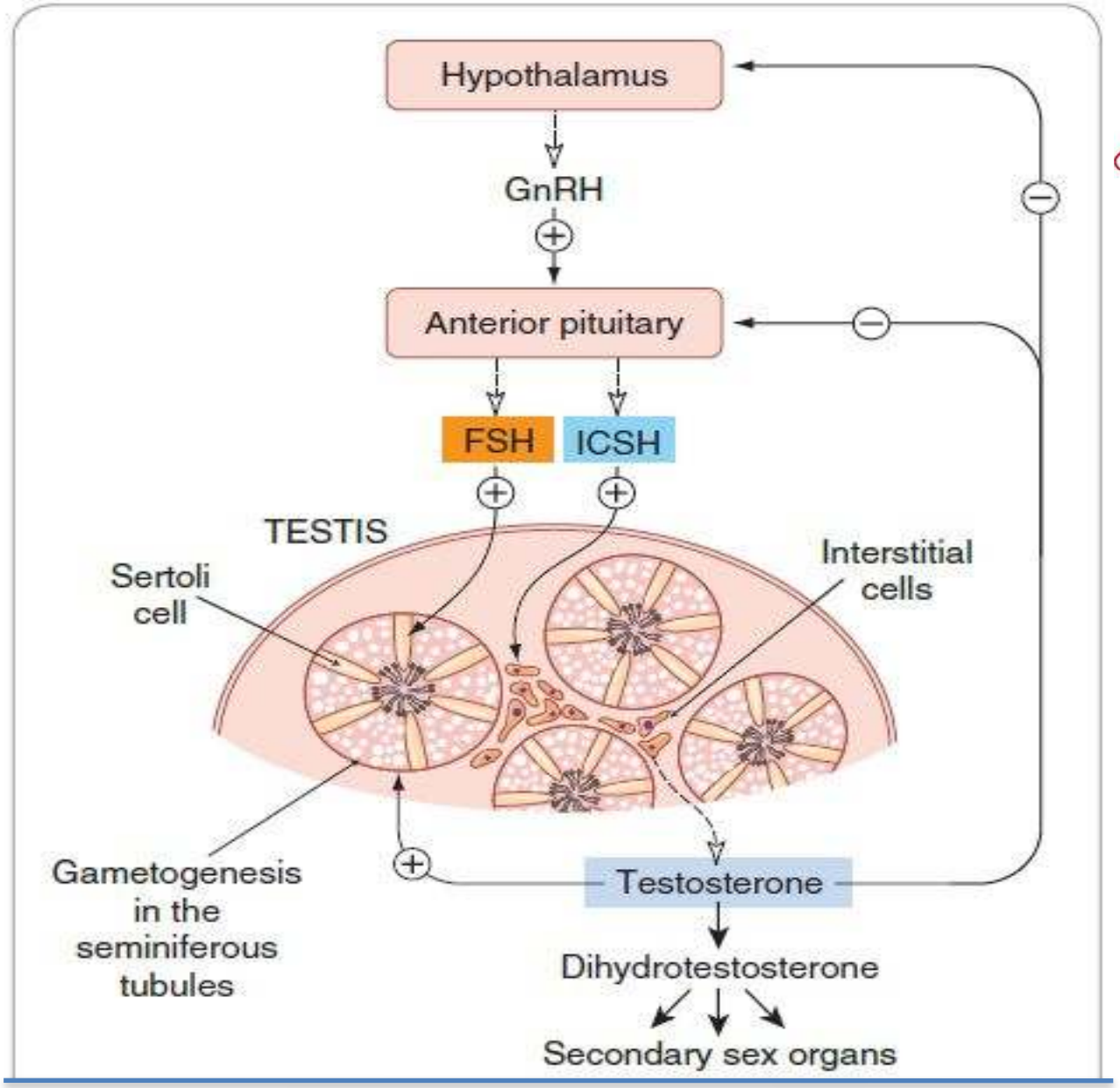
- Ovulation disorders
- Tubal abnormality
- Endometriosis

Male:

- Azospermia (impaired spermatogenesis or surgical sterilization)
- Oligospermia and impaired sperm motility ex: (varicocele)



Hormonal control of the female reproductive system



Hormonal control of the male reproductive system

Ovulation inducing drugs



1. Anti-estrogens:

Clomiphene citrate (Clomid)

Mechanism of action:

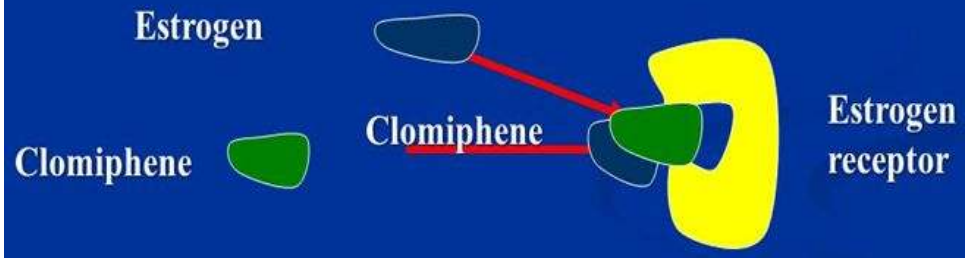
- ***Competitive antagonists*** blocks estrogen receptors in hypothalamus and pituitary → ↑ FSH and LH → stimulate ovulation.

Therapeutic uses:

- It is used to **stimulate ovulation** in many patients with infertility caused by ***lack of ovulation*** like:

1. ***polycystic ovarian syndrome(PCO)***, and
2. ***dysfunctional bleeding with an-ovulatory cycles.***
3. ***Diagnostic(*Clomiphene citrate challenge test CCCT)**

Clomiphene: Mechanism of Action



Structurally like estrogen, it binds to brain receptors where estrogen normally attaches, fooling body into thinking E2 is too low

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Pharmacokinetics:

- its active when taken orally
- undergoes enterohepatic circulation
- excreted in the faeces within 5 days after administration.

Adverse effects:

- Ovarian enlargement.
- hot flushes.
- increases the risk of multiple gestation, usually twins.

2. Aromatase inhibitors (AIs):



Aromatase inhibitors are a new class of oral estrogen synthesis inhibitors.

- **Anastrozole** and **letrozole** (Femara)

Mechanism of action:

- are non-steroidal reversible competitive inhibitors of aromatase which converts androstenedione and testosterone to oestrogens.

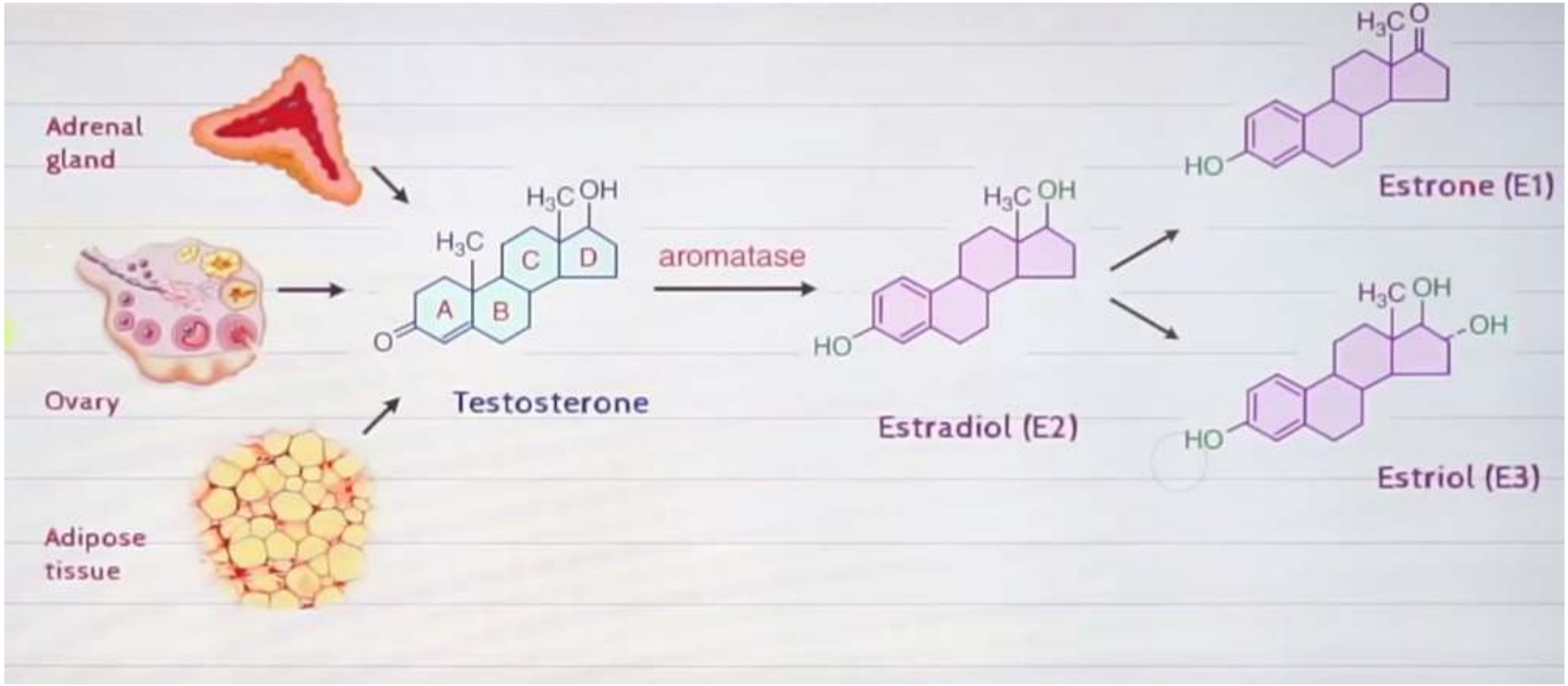
Therapeutic uses:

1) First and second line treatment of postmenopausal women with estrogen-receptor positive breast cancer
2) Adjuvant hormonal therapy of postmenopausal women with estrogen-receptor positive breast cancer.
3) Infertility due to anovulatory cycle

Given orally 99.9% bioavailability,

Side effects:

Hot flushes, hair loss, pain in joints and muscles, osteoporosis, difficulty sleeping, nausea and diarrhoea, depression.



The biosynthetic pathway for the androgens and oestrogens

3. Gonadotropins:

- Luteinizing hormone (LH)
- Follicle-stimulating hormone (FSH) and
- Human chorionic gonadotropin (hCG) *The chorion of placenta secretes human chorionic gonadotrophin (HCG), which maintains the lining of the uterus during pregnancy. HCG has an additional pharmacological action, used therapeutically in treating infertility, for stimulating ovulation*.

All three hormone FSH, LH, and hCG are ***glycoprotein*** that share an identical α chain in addition to a distinct β chain that confers receptor specificity of action. The β chains of hCG and LH are comparable, The action FSH are mediated by the FSH receptor, while those of LH and hCG are mediated by the LH receptor, and these two hormones are used interchangeably.

Preparations:

A. Menotropins:

This purified extract of FSH and LH is known as **menotropins**, or human menopausal gonadotropins **hMG (***menopur, repronex)* **.**



B. Follicle-stimulating hormone

Three forms of purified FSH are available.
Urofollitropin, also known as uFSH extracted from the urine of postmenopausal women
Two recombinant forms of FSH **(rFSH)** are also available: **follitropin alfa(**GONAL-F) and **follitropin beta (***Follistim, Puregon)*



C. Luteinizing_hormone:

Lutropin: has only been approved for use in combination with follitropin alfa for stimulation of follicular development in infertile women with profound LH deficiency

D.Human chorionic gonadotropin(hCG):

is produced by the human placenta and excreted into the urine where it can be extracted and purified and its given by **intramuscular injection**

Choriogonadotropin alfa (rhCG): is a recombinant form of hCG is administered by subcutaneous injection.



Mechanism of action:

- The actions of LH and hCG are mediated by the LH receptor,
- while those of FSH are mediated by the FSH receptor.

Clinical uses of Gonadotropins:

1. Treatment of infertility caused by lack of ***ovulation*** as a result of ***hypopituitarism***, or following failure of treatment with ***clomiphene***
2. Used by IVF specialists to increase the numbers of mature follicles to enable eggs to be collected for in vitro fertilisation. For this use, ***gonadotrophin is usually administered after secretion of endogenous FSH and LH has been suppressed***
3. Gonadotrophins are used in male with infertility caused by a low sperm count as a result of hypogonadotrophic hypogo nadism and cryptorchidism.
4. Human chorionic gonadotropin(hCG) and Choriogonadotropin alfa (rhCG) **trigger ovulation**

Adverse effects:

the most serious complications are the

1) **Ovarian hyperstimulation syndrome:** It is characterized by ovarian enlargement, ascites, hydrothorax, and hypovolemia, sometimes resulting in shock.
2) **Multiple pregnancies**: The risk of multiple pregnancy is estimated to be 15-20%, whereas the percentage of multiple pregnancies in the general population is closer to 1%
3) **Hemoperitoneum** (from a ruptured ovarian cyst).
4) **Arterial thromboembolism.**

4. Gonadotropin releasing Hormone & Its Analogs

- Pulsatile secretion of gonadotropin-releasing hormone (GnRH) from the hypothalamus is essential for release of the gonadotropins follicle-stimulating hormone (FSH) and luteinizing hormone (LH) from the anterior pituitary.

- Synthetic GnRH is termed gonadorelin, both agonists and antagonists, have been synthesised.
- **Agonists** are **Buserelin, goserelin** and **nafarelin**
- **Antagonists** are **cetrorelix, elagolix**.

Clinical uses

1. Gonadotrophin-releasing hormone agonists, given by subcutaneous infusion in pulses to mimic physiological secretion of GnRH, stimulate gonadotrophin release and induce ovulation and spermatogenesis in hypothalamic hypogonadotropic hypogonadism.

2. GnRH agonists and antagonists are used by specialists in infertility treatment, not to stimulate ovulation (which is achieved using gonadotropin preparations) but used **to inhibit LH surge in infertility protocols** before administration of FSH and HCG (Controlled ovarian hyperstimulation).

3. Continuous use, by nasal spray or as depot preparations or implant , to cause gonadal suppression in various sex hormone-dependent conditions, including

1. Prostate cancers
2. breast cancers, and
3. endometriosis
4. prcocious puberty



Advers effects of GnRH analogues

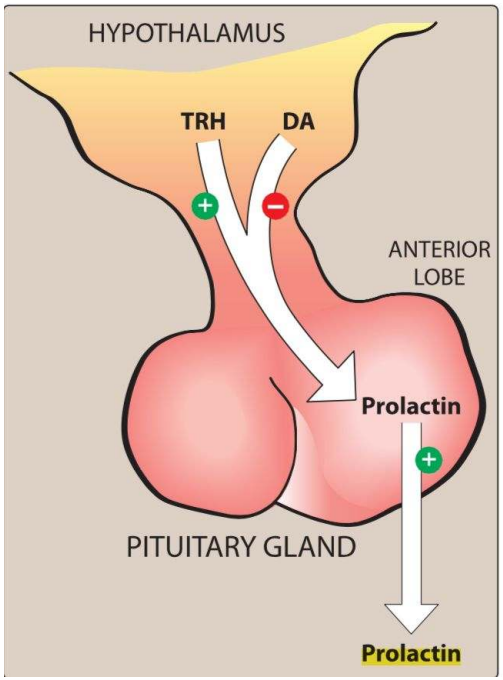
- Unwanted effects of GnRH agonists in women, for example ***flushing, vaginal dryness and bone loss***, ***depression, and ovarian cysts*** result from ***hypo-oestrogenism***
- The initial stimulation of gonadotrophin secretion on starting treatment can cause transient worsening of pain from bone metastases in men with prostate cancer, so treatment is started only after the patient has received an ***androgen receptor antagonist such as flutamide.***

4. Other drugs used for ovulation induction(Bromocriptine, cabergoline)

Hyperprolactinemia

is marked by abnormally high levels of the hormone prolactin in the blood. It is caused by

1. A benign tumor (prolactoma) in the pituitary gland.
2. Drugs that act as dopamine antagonists (for example, metoclopramide and some antipsychotics)



Secretion and action of prolactin. DA = dopamine; TRH = thyrotropin-releasing hormone.

Bromocriptine, cabergoline

D2 receptor agonists, inhibit prolactin production



- **Indication**:

1. galactorrhea and hypogonadism,
2. treatment of pituitary microadenomas.

- Given orally
- **Adverse effects** are nausea, vomiting, headache and postural hypotension.
- Cabergoline is more effective than bromocriptine in resolution of amenorrhea/oligomenorrhea due to galactorrhea with significantly fewer adverse events of nausea and vomiting.

Reference

- **Lippincott's Illustrated Reviews Pharmacology 6th edition**